

The Chain Rule & Directional Derivatives



The multivariable chain rule

- 1 Let $z = x^2 + y^2$, $x = \cos t$, $y = \sin t$. Find $\frac{dz}{dt}$.
 - 2 Let $w = xy + z^2$, $x = t$, $y = t^2$, $z = t^3$. Find $\frac{dw}{dt}$ at $t = 1$.
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Chain rule with two independent variables

- 3 Let $z = f(x, y)$ where $x = s + t$, $y = s - t$. Express $\frac{\partial z}{\partial s}$ in terms of $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$.
 - 4 For $z = x^2y$, $x = s + t$, $y = s - t$, find $\frac{\partial z}{\partial s}$ and $\frac{\partial z}{\partial t}$ at $s = 2$, $t = 1$.
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Gradient and directional derivative

- 5 Find the gradient of $f(x, y) = x^2y^3$ at $(1, 2)$.
 - 6 Find the directional derivative of $f(x, y) = x^2y$ at $(2, 1)$ in the direction of $\mathbf{v} = \langle 1, 1 \rangle$.
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The gradient points in the direction of steepest ascent

- 7 For $f(x, y) = 3x^2 - y^2$ at $(1, 2)$, find the direction of steepest ascent and the maximum rate of increase.
 - 8 In what direction from $(1, 1)$ does $f(x, y) = xy$ decrease most rapidly, and at what rate?
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Application of the gradient

- 9 The temperature on a plate is $T(x, y) = 100 - x^2 - 2y^2$. A bug at $(2, 1)$ wants to cool off fastest. In which direction should it move?
- 10 Find the rate of change of $T(x, y) = 100 - x^2 - 2y^2$ at $(2, 1)$ in the direction of the unit vector $\langle \frac{3}{5}, \frac{4}{5} \rangle$.